JEE Mains 2019 Chapter wise Question Bank

Trigonometrical Equations – Questions

Q1

If $0 \le x < \frac{\pi}{2}$, then the number of values of x for which $\sin x - \sin 2x + \sin 3x = 0$, is:

(1) 3 (2) 1
(3) 4 (4) 2

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Q2

The sum of all values of $\theta \in \left(0, \frac{\pi}{2}\right)$ satisfying $\sin^2 2\theta + \cos^4 2\theta = \frac{3}{4}$ is:

(1) π (2) $\frac{5\pi}{4}$ (3) $\frac{\pi}{2}$ (4) $\frac{3\pi}{8}$

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Q3

All the pairs (x, y) that satisfy the inequality $2\sqrt{\sin^2 x - 2\sin x + 5}.\frac{1}{4\sin^2 y} \le 1$ also satisfy the equation:

(1) $2|\sin x| = 3\sin y$ (2) $2 \sin x = \sin y$
(3) $\sin x = 2 \sin y$ (4) $\sin x = |\sin y|$

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Q4

The number of solutions of the equation $1 + \sin^4 x = \cos^2 3x, x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is :

(1) 3 (2) 5 (3) 7 (4) 4

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Q5

Let S be the set of all $\alpha \in R$ such that the equation, $\cos 2x + \alpha \sin x = 2\alpha - 7$ has a solution. Then S is equal to :

(1) R (2) [1, 4] (3) [3,7] (4) [2, 6]

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Q6

A value of $\theta \in (0, \pi/3)$, for which

$$\begin{vmatrix} 1+\cos^2\theta & \sin^2\theta & 4\cos 6\theta \\ \cos^2\theta & 1+\sin^2\theta & 4\cos 6\theta \\ \cos^2\theta & \sin^2\theta & 1+4\cos 6\theta \end{vmatrix} = 0, \text{ is :}$$

(1) $\frac{\pi}{9}$ (2) $\frac{\pi}{18}$ (3) $\frac{7\pi}{24}$ (4) $\frac{7\pi}{36}$

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Q7

If $[x]$ denotes the greatest integer $\le x$, then the system of linear equations

$[\sin\theta]\, x + [-\cos\theta]\, y = 0$

$[\cot\theta]\, x + y = 0$

(1) have infinitely many solutions if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and has a unique solution if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$.

(2) has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$.

(3) has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and have infinitely many solutions if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$.

(4) have infinitely many solutions if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

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Trigonometrical Equations - Solutions

Q1

(4) $\sin x - \sin 2x + \sin 3x = 0$

$\Rightarrow \sin x - 2\sin x.\cos x + 3\sin x - 4\sin^3 x = 0$

$\Rightarrow 4\sin x - 4\sin^3 x - 2\sin x.\cos x = 0$

$\Rightarrow 2\sin x(1 - \sin^2 x) - \sin x.\cos x = 0$

$\Rightarrow 2\sin x.\cos^2 x - \sin x.\cos x = 0$

$\Rightarrow \sin x.\cos x(2\cos x - 1) = 0$

$\therefore \quad \sin x = 0, \cos x = 0, \cos x = \frac{1}{2}$

$\therefore \quad x = 0, \frac{\pi}{3} \qquad \because \quad x \in \left[0, \frac{\pi}{2}\right)$

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Q2

(3) $\sin^2 2\theta + \cos^4 2\theta = \frac{3}{4}$

$\Rightarrow 1 - \cos^2 2\theta + \cos^4 2\theta = \frac{3}{4}$

$\Rightarrow \cos^2 2\theta(1 - \cos^2 2\theta) = \frac{1}{4}$ ...(1)

$\because$ G.M. $\leq$ A.M.

$\therefore \quad (\cos^2 2\theta)(1 - \cos^2 2\theta) \leq$

$$\left(\frac{\cos^2 2\theta + (1 - \cos^2 2\theta)}{2}\right)^2$$

$= \frac{1}{4}$ ...(2)

So, from eq. (1) & (2), we get

G.M. = A.M.

It is possible only if,

$$\cos^2 2\theta = 1 - \cos^2 2\theta$$

$\Rightarrow \cos^2 2\theta = \frac{1}{2}$

$\Rightarrow \cos 2\theta = \pm\frac{1}{\sqrt{2}}$

$\Rightarrow \theta = \frac{\pi}{8}, \frac{3\pi}{8}$

$\therefore \quad$ Sum $= \frac{\pi}{8} + \frac{3\pi}{8} = \frac{\pi}{2}$

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Q3

(4) Given inequality is,

$$2\sqrt{\sin^2 x - 2\sin x + 5} \le 2^{2\sin^2 y}$$

$$\Rightarrow \sqrt{\sin^2 x - 2\sin x + 5} \le 2\sin^2 y$$

$$\Rightarrow \sqrt{(\sin x - 1)^2 + 4} \le 2\sin^2 y$$

It is true if $\sin x = 1$ and $|\sin y| = 1$

Therefore, $\sin x = |\sin y|$

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Q4

(2) Consider equation, $1 + \sin^4 x = \cos^2 3x$

L.H.S. $= 1 + \sin^4 x$ and R.H.S. $= \cos^2 3x$

$\because$ L.H.S. ≥ 1 and R.H.S. $\le 1 \Rightarrow$ L.H.S. = R.H.S. = 1

$\sin^4 x = 0$, and $\cos^2 3x = 1$

$\Rightarrow \sin x = 0$ and $(4\cos^2 x - 3)^2 \cos^2 x = 1$

$\Rightarrow \sin x = 0$ and $\cos^2 x = 1 \Rightarrow x = 0, \pm\pi, \pm 2\pi$

Hence, total number of solutions is 5.

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Q5

(4) Given equation is, $\cos 2x + \alpha \sin x = 2\alpha - 7$

$1 - 2\sin^2 x + \alpha \sin x = 2\alpha - 7$

$2\sin^2 x - \alpha \sin x + (2\alpha - 8) = 0$

$$\Rightarrow \sin x = \frac{\alpha \pm \sqrt{\alpha^2 - 8(2\alpha + 8)}}{4}$$

$$\Rightarrow \sin x = \frac{\alpha \pm (\alpha - 8)}{4} \quad \Rightarrow \sin x = \frac{\alpha - 4}{4}$$

[$\sin x = 2$ (rejected)]

$\because$ equation has solution, then $\dfrac{\alpha - 4}{4} \in [-1, 1]$

$\Rightarrow \alpha \in [2, 6]$

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Q6

(1) $C_1 \to C_1 + C_2$

$$\begin{vmatrix} 2 & \sin^2\theta & 4\cos 6\theta \\ 2 & 1+\sin^2\theta & 4\cos 6\theta \\ 1 & \sin^2\theta & 1+4\cos 6\theta \end{vmatrix} = 0$$

$R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$

$$\begin{vmatrix} 0 & -1 & 0 \\ 1 & 1 & -1 \\ 1 & \sin^2\theta & (1+4\cos 6\theta) \end{vmatrix} = 0$$

On expanding, we get $2 + 4\cos 6\theta = 0$

$$\cos 6\theta = -\frac{1}{2} \quad \because \theta \in \left(0, \frac{\pi}{3}\right) \Rightarrow 6\theta \in (0, 2\pi)$$

Therefore, $6\theta = \dfrac{2\pi}{3}$ or $\dfrac{4\pi}{3} \Rightarrow \theta = \dfrac{\pi}{9}$ or $\dfrac{2\pi}{9}$

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Q7

(1) According to the question, there are two cases.

Case 1 : $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

In this interval, $[\sin\theta] = 0$, $[-\cos\theta] = 0$ and $[\cot\theta] = -1$

Then the system of equations will be ;

$0 \cdot x + 0 \cdot y = 0$ and $-x + y = 0$

Which have infinitely many solutions.

Case 2 : $\theta \in \left(\pi, \frac{7\pi}{6}\right)$

In this interval, $[\sin\theta] = -1$ and $[-\cos\theta] = 0$,

Then the system of equations will be ;

$-x + 0 \cdot y = 0$ and $[\cot\theta]\, x + y = 0$

Clearly, $x = 0$ and $y = 0$ which has unique solution.

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